Do Now · retrieval

Linked lists with arrays and pointers

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the main difference between an array and a list?

Q2

In an array, what does an index identify?

Q3

In a linked list, what does null in a node's next pointer mean?

Q4

If the first item is at index 4 and its next pointer is 1, where is the second item?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start pointer — CODE:
int start = 2; — The first logical item is stored at index 2, not necessarily index 0.

Core idea 2

Follow next pointers — CODE:

```
int current = start;

while (current != -1)
{
Console.WriteLine(data[current]);
current = next[current];
```

} — Use -1 as null in code. Each pointer decides the next index.

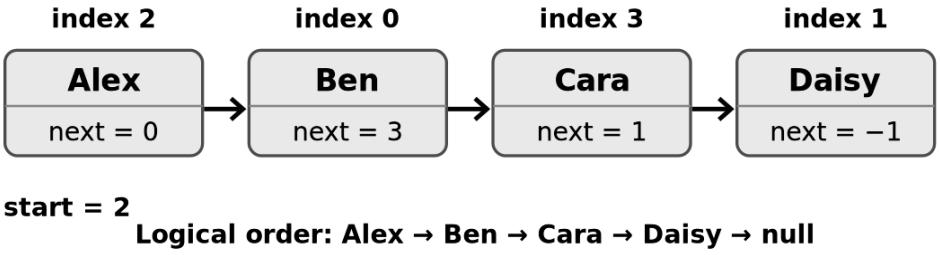
Core idea 3

Insert at the front — CODE:

```
next[newIndex] = start;
```

start = newIndex; — The new item points to the old first item, then start changes.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Remove after current — CODE:
next[current] = next[next[current]]; — The pointer bypasses one node. The array data itself does not need to shift.

Vocabulary · use the correct technical terms

linked list — A structure where each item contains data and a link to another item.

pointer — A value identifying the location of another item.

start pointer — The index of the first item in the list.

next pointer — The index of the next item.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Traverse Alex → Ben → Cara → Daisy

Follow the worked method

1. Use the start pointer — CODE:
current = 2
data[2] = "Alex"

2. Continue pointers — CODE:
next[0] = 3
data[3] = "Cara"

next[3] = 1
data[1] = "Daisy"

3. Reach null — CODE:
next[1] = -1
stop

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the same trace
CODE:
firstIndex = start
secondIndex = ?
secondData = data[?]

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Identify the first logical item
Run the starter. State the start index and first item. Explain why index 0 is not necessarily the head.
CODE:

```
using System;
public class Program
{
public static void Main()
{
string[] data = {"Ben", "Daisy", "Alex", "Cara", "", ""};
int[] next = {3, -1, 0, 1, -1, -1};
int start = 2; // slots 4 and 5 are unused
Console.WriteLine(data[start]);
}
}
```

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Combine, test and explain
Combine traversal, insertion and deletion using the pointer arrays. Record expected and actual outputs for the tests below.
CODE:
Empty list: start=-1
One node: next[start]=-1
Insert into a spare slot
Delete head and middle nodes
Reject insertion if no slot is free
Use -1 only as the null marker. Reuse a removed slot only after marking it free in your chosen allocation scheme.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In an array-based linked list, the item at the next array index is not necessarily the next logical item.

Use key vocabulary from memory

1. Explain linked list in your own words.

2. Explain pointer in your own words.

3. Explain start pointer in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.